

House Price Prediction – Project Summary

Introduction

The House Price Prediction project was developed to estimate residential property prices using Machine Learning techniques. In the real estate industry, determining the correct price of a property is often challenging because multiple factors such as area, number of bedrooms, bathrooms, stories, parking facilities, and furnishing status influence the final price. The objective of this project is to build an intelligent regression model that can accurately predict house prices based on these features and help buyers, sellers, and real estate businesses make informed decisions.

Data Preparation and Preprocessing

The project utilized the Housing Prices Dataset obtained from Kaggle. Initially, the dataset was loaded using the Pandas library and explored to understand its structure, dimensions, and data types. Missing values and duplicate records were checked and handled appropriately to ensure data quality. Since several attributes contained categorical values such as "yes" and "no", these variables were converted into numerical format using one-hot encoding. This preprocessing step made the dataset suitable for machine learning algorithms and improved the overall model performance.

Model Development

Two regression models were implemented and compared in this project: Linear Regression and Random Forest Regressor. The dataset was divided into training and testing sets using an 80:20 ratio. Linear Regression was selected as a baseline model because of its simplicity and ability to identify linear relationships between variables. Random Forest Regressor was chosen because it can capture complex non-linear patterns and interactions among features. Both models were trained on the training dataset and evaluated on the testing dataset.

Performance Evaluation

The performance of the models was measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score. These evaluation metrics provide insights into prediction accuracy and model reliability. The Random Forest Regressor generally produced better results compared to Linear Regression due to its ensemble learning capability. A higher R^2 score indicated that the model successfully explained a large portion of the variance in house prices, while lower MAE and RMSE values demonstrated reduced prediction errors.

Data Visualization and Insights

Several visualizations were created to better understand the dataset and model performance. A histogram of house prices showed the overall distribution of property values. A correlation heatmap

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revealed the relationships between different housing features and the target variable. An Actual vs Predicted Price scatter plot was generated to visualize how closely the predicted values matched the actual house prices. The analysis revealed that area, number of bathrooms, stories, parking availability, and air conditioning facilities were among the most influential factors affecting house prices.

Conclusion and Recommendation

The project successfully demonstrated the application of machine learning in the real estate domain. Among the implemented models, Random Forest Regressor achieved superior prediction accuracy and provided more reliable estimates of house prices. The findings indicate that larger properties with better facilities and infrastructure tend to command higher market values. Real estate businesses can leverage such predictive models to establish fair pricing strategies, improve market analysis, and support customers in making data-driven property investment decisions. Future improvements may include incorporating additional location-based factors, economic indicators, and advanced machine learning algorithms to further enhance prediction accuracy.